

The 3-qubit, 2-cavity Hamiltonian

Matched $\chi_{qA} = \chi_{qB}$ JP gate and unitless Lindblad rates

qumode_sep technical note

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Abstract

Laboratory Hamiltonian of two storage cavities A, B and three transmons d, q, e ; conversion of device frequencies into the unitless rates used by a Lindblad solver; and the joint-parity (JP) gate as a finite-time evolution under a matched shared dispersive coupling.

1 Hardware

Five modes in a chain $d-A-q-B-e$ (Fig. 1). Cavities A, B are the qumodes. Transmons d, e are local ECD ancillas. Transmon q is the shared coupler. This is the two-cavity extension of the one-qumode device in Dutta *et al.* [1] and the Alice/Bob layout of Gao *et al.* [3].

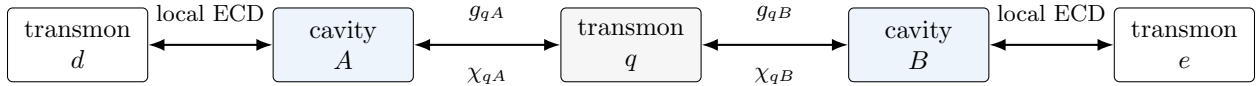


Figure 1: Three transmons and two cavities. For a single-wait JP we set $\chi_{qA} = \chi_{qB}$.

2 Laboratory Hamiltonian

Let $\hat{a}$ ($\hat{b}$) annihilate a photon in A (B), and let $\hat{c}_d, \hat{c}_q, \hat{c}_e$ lower the transmons. Rotating-wave form:

$$\begin{aligned}
 \frac{\hat{H}}{\hbar} = & \omega_A \hat{a}^\dagger \hat{a} + \omega_B \hat{b}^\dagger \hat{b} + \sum_{j \in \{d, q, e\}} \left[\omega_j \hat{c}_j^\dagger \hat{c}_j + \frac{\alpha_j}{2} \hat{c}_j^\dagger \hat{c}_j (\hat{c}_j^\dagger \hat{c}_j - 1) \right] \\
 & + g_{dA} (\hat{a}^\dagger \hat{c}_d + \hat{a} \hat{c}_d^\dagger) + g_{eB} (\hat{b}^\dagger \hat{c}_e + \hat{b} \hat{c}_e^\dagger) \\
 & + g_{qA} (\hat{a}^\dagger \hat{c}_q + \hat{a} \hat{c}_q^\dagger) + g_{qB} (\hat{b}^\dagger \hat{c}_q + \hat{b} \hat{c}_q^\dagger) + \hat{H}_{\text{drive}}(t)/\hbar.
 \end{aligned} \tag{1}$$

Direct $d-B$, $e-A$, and $d-e$ couplings are dropped. Cavities are detuned by ~ 1 GHz so they do not hybridize [3].

Far off resonance, $\Delta_{j\mu} = \omega_j - \omega_\mu$, each transmon-cavity pair has a dispersive shift $\chi_{j\mu} = g_{j\mu}^2 \alpha_j / [\Delta_{j\mu} (\Delta_{j\mu} + \alpha_j)]$.

In the lab, χ_{qA} is a measured frequency difference, not a derived symbol. Probe cavity A with a weak tone: the resonance peak sits at one frequency when q is in $|g\rangle$ and at another when q is in $|e\rangle$. That peak-to-peak gap is $\chi_{qA}/2\pi$ (typically quoted in MHz). Equivalently, park a photon in A and spectroscopy q : the transmon line splits into a ladder $\omega_q, \omega_q - \chi_{qA}, \omega_q - 2\chi_{qA}, \dots$ indexed

by photon number n_A (number splitting). Either way, what moved is a resonance you read on a network analyzer or a qubit spectroscopy scan. The same measurement on q - B , d - A , and e - B gives χ_{qB} , χ_{dA} , and χ_{eB} .

$$\begin{aligned} \frac{\hat{H}_{\text{disp}}}{\hbar} = & \tilde{\omega}_A \hat{n}_A + \tilde{\omega}_B \hat{n}_B + \sum_{j \in \{d, q, e\}} \tilde{\omega}_j |e\rangle\langle e|_j \\ & - \chi_{dA} \hat{n}_A |e\rangle\langle e|_d - \chi_{eB} \hat{n}_B |e\rangle\langle e|_e - \chi_{qA} \hat{n}_A |e\rangle\langle e|_q - \chi_{qB} \hat{n}_B |e\rangle\langle e|_q \\ & - \frac{K_A}{2} \hat{n}_A (\hat{n}_A - 1) - \frac{K_B}{2} \hat{n}_B (\hat{n}_B - 1) - K_{AB} \hat{n}_A \hat{n}_B, \end{aligned} \quad (2)$$

with $\hat{n}_A = \hat{a}^\dagger \hat{a}$, $\hat{n}_B = \hat{b}^\dagger \hat{b}$. The $\tilde{\omega}$ are Lamb-shifted bare frequencies. They imprint ordinary dynamical phases $e^{-i\tilde{\omega}_A \tau}$ on Fock states; Kerr imprints extra n -dependent phases. Both are kept in the Lindblad generator below.

Matched shared coupling. JP is a wait under $-\chi_{qA} \hat{n}_A |e\rangle\langle e|_q - \chi_{qB} \hat{n}_B |e\rangle\langle e|_q$. One wait of duration τ maps joint parity onto q iff $\chi_{qA}\tau$ and $\chi_{qB}\tau$ are both odd multiples of π . The easy choice is to set them equal,

$$\chi_{qA} = \chi_{qB}, \quad \tau = \pi / \chi_{qA}. \quad (3)$$

No extra symbol is needed: after this identification the shared term is $-\chi_{qA}(\hat{n}_A + \hat{n}_B) |e\rangle\langle e|_q$. Because $\omega_A \neq \omega_B$, equal $\chi_{qA} = \chi_{qB}$ generally means *unequal* g_{qA}, g_{qB} . Unmatched shared shifts (Wang: 0.71 vs 1.41 MHz [4]) need two different waits; we do not use that here.

The local shifts χ_{dA} and χ_{eB} are *not* used by JP. They set the ECD rate on (d, A) and (e, B) and may be chosen independently of χ_{qA} and of each other (Eickbusch ECD: $\chi_{dA}/2\pi = 33$ kHz while $\chi_{qA}/2\pi \sim 1$ MHz). They still act during the JP wait as spectators $-\chi_{dA} \hat{n}_A |e\rangle\langle e|_d - \chi_{eB} \hat{n}_B |e\rangle\langle e|_e$. If d or e is in superposition those terms add extra phases; echo $X_d X_e$ at the midpoint, or leave them in the Lindblad Hamiltonian and accept the error.

3 From SI frequencies to unitless numbers

A Lindblad solver takes $\hbar = 1$ and a Hamiltonian whose eigenvalues are inverse times. Pick one laboratory rate as the unit. Here that unit is the matched shared shift χ_{qA} (with $\chi_{qA} = \chi_{qB}$, so $\chi_{qA} = 1$ in code).

Recipe. For every laboratory angular frequency Ω (already in rad/s, i.e. 2π times the quoted Hz), store

$$\tilde{\Omega} = \Omega / \chi_{qA}. \quad (4)$$

Laboratory time t becomes $\tilde{t} = \chi_{qA} t$. Jump rates follow the same rule: $\tilde{\kappa} = \kappa / \chi_{qA}$. Dimensionless noise over a gate of duration τ is the product $\kappa\tau = \tilde{\kappa} \tilde{\tau}$, which is the parameter used by Dutta *et al.* [1].

From Eickbusch [2] and Wang [4]:

quantity	laboratory	unit $\chi_{qA}/2\pi = 1$ MHz	notes
$\chi_{qA} = \chi_{qB}$	1 MHz (choose matched)	1	time unit $1/\chi_{qA} \approx 0.16 \mu\text{s}$
$\chi_{dA} = \chi_{eB}$	33 kHz [2]	0.033	local ECD, kept weak
$\kappa_A = \kappa_B = 1/T_{1,\text{cav}}$	$T_{1,\text{cav}} = 436 \mu\text{s}$	$\tilde{\kappa} = 3.65 \times 10^{-4}$	$\kappa/2\pi \simeq 365$ Hz [1]
$\gamma_q = 1/T_{1,q}$	$T_{1,q} \approx 50 \mu\text{s}$	$\tilde{\gamma}_q = 3.18 \times 10^{-3}$	transmon relaxation
$\gamma_{\phi,q}$ from $T_2 \approx 30 \mu\text{s}$	$T_2^{-1} - T_1^{-1}/2$	$\tilde{\gamma}_{\phi,q} \approx 1.8 \times 10^{-3}$	extra dephasing of q

Conversion of the cavity row, written out:

$$\chi_{qA} = 2\pi \times 10^6 \text{ s}^{-1}, \quad \kappa = \frac{1}{436 \text{ } \mu\text{s}} \simeq 2.29 \times 10^3 \text{ s}^{-1}, \quad \tilde{\kappa} = \frac{\kappa}{\chi_{qA}} = 3.65 \times 10^{-4}. \quad (5)$$

A wait $\tau = \pi/\chi_{qA}$ then has $\kappa\tau = \pi\tilde{\kappa} \simeq 1.15 \times 10^{-3}$, the same 10^{-3} scale as the per-ECD-block $\kappa\tau \sim 1.5 \times 10^{-3}$ in Ref. [1].

4 JP gate

The joint-parity operator on the two cavities is

$$\hat{\Pi}_{AB} = (-1)^{\hat{n}_A + \hat{n}_B} = \hat{\Pi}_A \hat{\Pi}_B, \quad \hat{\Pi}_A = (-1)^{\hat{n}_A}, \quad \hat{\Pi}_B = (-1)^{\hat{n}_B}. \quad (6)$$

It is diagonal in the Fock basis. On a product number state $|n, m\rangle \equiv |n\rangle_A \otimes |m\rangle_B$,

$$\hat{\Pi}_{AB}|n, m\rangle = (-1)^{n+m}|n, m\rangle = \begin{cases} +|n, m\rangle & n+m \text{ even,} \\ -|n, m\rangle & n+m \text{ odd.} \end{cases} \quad (7)$$

Equivalently, the four parity sectors are

parity of n	parity of m	$n+m$	$\hat{\Pi}_{AB}$ eigenvalue
even	even	even	+1
even	odd	odd	-1
odd	even	odd	-1
odd	odd	even	+1

The odd-joint-parity projector is the +1 eigenspace of $(\hat{\mathbb{I}} - \hat{\Pi}_{AB})/2$:

$$\hat{P}_{\text{odd}} = \frac{\hat{\mathbb{I}} - \hat{\Pi}_{AB}}{2}, \quad \hat{P}_{\text{odd}}|n, m\rangle = \begin{cases} 0 & n+m \text{ even,} \\ |n, m\rangle & n+m \text{ odd.} \end{cases} \quad (8)$$

The JP gate is a $\pi/2$ phase on that subspace,

$$\boxed{\hat{U}_{\text{JP}} = \exp[-i(\pi/2)\hat{P}_{\text{odd}}] = e^{-i\pi/4} \exp[+i(\pi/4)\hat{\Pi}_{AB}]}, \quad (9)$$

so

$$\hat{U}_{\text{JP}}|n, m\rangle = (-i)^{(n+m) \bmod 2}|n, m\rangle = \begin{cases} +|n, m\rangle & n+m \text{ even,} \\ -i|n, m\rangle & n+m \text{ odd.} \end{cases} \quad (10)$$

Time evolution that realizes it. Initialize q in $|g\rangle$. With $\chi_{qA} = \chi_{qB}$ the shared term in Eq. (2) is $\hat{H}_{\text{JP}}/\hbar = -\chi_{qA}(\hat{n}_A + \hat{n}_B)|e\rangle\langle e|_q$. In unitless time ($\chi_{qA} = 1$), $\hat{\tilde{H}}_{\text{JP}} = -(\hat{n}_A + \hat{n}_B)|e\rangle\langle e|_q$, so a wait $\tilde{\tau} = \pi$ is

$$\hat{U}_\pi = |g\rangle\langle g|_q + |e\rangle\langle e|_q \hat{\Pi}_{AB} : \quad \hat{U}_\pi|g, n, m\rangle = |g, n, m\rangle, \quad \hat{U}_\pi|e, n, m\rangle = (-1)^{n+m}|e, n, m\rangle. \quad (11)$$

Qubit rotations are $R_y(\pi/2)|g\rangle = (|g\rangle + |e\rangle)/\sqrt{2}$ and $R_y(-\pi/2)(|g\rangle \pm |e\rangle)/\sqrt{2} = |g\rangle$ or $|e\rangle$ (even/odd superposition). Virtual $S^\dagger|g\rangle = |g\rangle$, $S^\dagger|e\rangle = -i|e\rangle$. Write $\lambda = (-1)^{n+m}$ and start from $|g, n, m\rangle$.

Even ($n + m$ even, $\lambda = +1$):

$$\begin{aligned}
R_y(+\pi/2): & (|g\rangle + |e\rangle)/\sqrt{2} |n, m\rangle \\
\hat{U}_\pi: & (|g\rangle + |e\rangle)/\sqrt{2} |n, m\rangle \quad (\text{no relative phase}) \\
R_y(-\pi/2): & |g, n, m\rangle \\
S^\dagger: & |g, n, m\rangle \\
R_y(+\pi/2), \hat{U}_\pi, R_y(-\pi/2): & |g, n, m\rangle.
\end{aligned}$$

Net: $+|n, m\rangle$, q back in $|g\rangle$.

Odd ($n + m$ odd, $\lambda = -1$):

$$\begin{aligned}
R_y(+\pi/2): & (|g\rangle + |e\rangle)/\sqrt{2} |n, m\rangle \\
\hat{U}_\pi: & (|g\rangle - |e\rangle)/\sqrt{2} |n, m\rangle \\
R_y(-\pi/2): & -|e, n, m\rangle \quad (\text{parity now stored on } q) \\
S^\dagger: & i|e, n, m\rangle \quad (-i \text{ on } |e\rangle \text{ times the minus sign}) \\
R_y(+\pi/2): & i(-|g\rangle + |e\rangle)/\sqrt{2} |n, m\rangle \\
\hat{U}_\pi: & i(-|g\rangle - |e\rangle)/\sqrt{2} |n, m\rangle \\
R_y(-\pi/2): & -i|g, n, m\rangle.
\end{aligned}$$

Net: $-i|n, m\rangle$, q back in $|g\rangle$.

Together this is $|g\rangle \otimes \hat{U}_{\text{JP}}|n, m\rangle$. Local d, e are spectators (echo them if χ_{dA}, χ_{eB} must not act).

Duration. Idle time is two waits of π/χ_{qA} . With $\chi_{qA}/2\pi = 1$ MHz,

$$\tau_{\text{JP}} = \frac{2\pi}{\chi_{qA}} = 1 \text{ } \mu\text{s} \quad (\tilde{\tau}_{\text{JP}} = 2\pi \text{ in code}). \quad (12)$$

Four transmon $\pi/2$ pulses add ~ 100 ns ($t_q = 24$ ns [2]) and can be treated as instantaneous relative to the waits.

Lindblad equation during the waits (unitless, $\chi_{qA} = \chi_{qB} = 1$). Use the *full* 3-qubit, 2-cavity Hamiltonian Eq. (2) so every Fock- and transmon-dependent phase is present:

$$\begin{aligned}
\hat{\tilde{H}} = \frac{\hat{H}_{\text{disp}}}{\hbar\chi_{qA}} = & \tilde{\omega}_A \hat{n}_A + \tilde{\omega}_B \hat{n}_B + \sum_{j \in \{d, q, e\}} \tilde{\omega}_j |e\rangle\langle e|_j \\
& - \tilde{\chi}_{dA} \hat{n}_A |e\rangle\langle e|_d - \tilde{\chi}_{eB} \hat{n}_B |e\rangle\langle e|_e - (\hat{n}_A + \hat{n}_B) |e\rangle\langle e|_q \\
& - \frac{\tilde{K}_A}{2} \hat{n}_A (\hat{n}_A - 1) - \frac{\tilde{K}_B}{2} \hat{n}_B (\hat{n}_B - 1) - \tilde{K}_{AB} \hat{n}_A \hat{n}_B,
\end{aligned} \quad (13)$$

with $\tilde{\chi}_{dA} = \tilde{\chi}_{eB} = 0.033$ and $\tilde{K} \sim 10^{-3}$ – 10^{-2} ($K/2\pi \sim 1$ – 10 kHz [4]). The $\tilde{\omega}$ rows are the free phases ($\tilde{\omega}_A = \tilde{\omega}_A/\chi_{qA} \simeq 5260$ if the cavity is at 5.26 GHz). They can be moved into an interaction picture $\hat{\tilde{H}}_I = e^{i\hat{\tilde{H}}_0 t} (\hat{\tilde{H}} - \hat{\tilde{H}}_0) e^{-i\hat{\tilde{H}}_0 t}$ with $\hat{\tilde{H}}_0 = \tilde{\omega}_A \hat{n}_A + \tilde{\omega}_B \hat{n}_B + \sum_j \tilde{\omega}_j |e\rangle\langle e|_j$, which removes the stiff GHz numbers without dropping Kerr or any $\chi_{j\mu}$. Either picture is equivalent; the interaction picture is what a Lindblad integrator should use.

Then

$$\frac{d\hat{\rho}}{dt} = -i[\hat{\tilde{H}}, \hat{\rho}] + \tilde{\kappa} \mathcal{D}[\hat{a}]\hat{\rho} + \tilde{\kappa} \mathcal{D}[\hat{b}]\hat{\rho} + \sum_{j \in \{d, q, e\}} \left(\tilde{\gamma}_j \mathcal{D}[\hat{\sigma}_-^{(j)}]\hat{\rho} + \frac{\tilde{\gamma}_{\phi, j}}{2} \mathcal{D}[\hat{\sigma}_z^{(j)}]\hat{\rho} \right), \quad (14)$$

with $\mathcal{D}[\hat{L}]\hat{\rho} = \hat{L}\hat{\rho}\hat{L}^\dagger - \frac{1}{2}\{\hat{L}^\dagger\hat{L}, \hat{\rho}\}$, using $\hat{H}$ or $\hat{H}_I$ as above. The $(\hat{n}_A + \hat{n}_B)|e\rangle\langle e|_q$ term implements JP; the local χ_{dA}, χ_{eB} and Kerr terms add extra phases on $|n, m\rangle$ and on d, e . Integrate each wait from $\tilde{t} = 0$ to π ($\tilde{\kappa} = 3.65 \times 10^{-4}$, $\tilde{\gamma}_d = \tilde{\gamma}_q = \tilde{\gamma}_e = 3.18 \times 10^{-3}$). Cavity loss over the full JP is $\kappa\tau_{\text{JP}} \simeq 2.3 \times 10^{-3}$; transmon T_1 while q spends half the gate in $|e\rangle$ is the larger error.

References

- [1] R. Dutta *et al.*, “Solving constrained optimization problems using hybrid qubit–qumode quantum devices,” arXiv:2501.11735. Defines the dimensionless noise parameter $\kappa\tau$; per ECD block $\kappa\tau \sim 1.5 \times 10^{-3}$ on the Eickbusch device.
- [2] A. Eickbusch *et al.*, *Nat. Phys.* **18**, 1464 (2022). Cavity 5.26 GHz, $T_{1,\text{cav}} = 436 \mu\text{s}$; transmon 6.65 GHz, $\alpha/2\pi = -181 \text{ MHz}$, $T_{1,q} \approx 50 \mu\text{s}$, $T_2 \approx 30 \mu\text{s}$; local $\chi_{dA}/2\pi = 33 \text{ kHz}$; $t_q = 24 \text{ ns}$.
- [3] Y. Y. Gao *et al.*, *Phys. Rev. X* **8**, 021073 (2018). Two cavities detuned by $\sim 1 \text{ GHz}$; local ancilla plus shared coupler.
- [4] C. Wang *et al.*, *Science* **352**, 1087 (2016). Unmatched shared shifts $\chi_A^{ge}/2\pi = 0.71 \text{ MHz}$, $\chi_B^{ge}/2\pi = 1.41 \text{ MHz}$ (we instead *match* them).